Setting up xv6 Compilation and using the OS Class Resources

xv6 a simple Unix-like teaching Operating System

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Cloning the CSci Operating Systems Class Repository

- You first need to have Git / Docker / VSCode DevContainers setup so you can clone repositories and create docker dev containers.
- Clone the csci530-os-xv6 Class Resources Repository
  - `git clone https://github.com/csci530-os/csci530-os-xv6`
  - Can only clone read only using `https` url
- Reopen container in a Docker / DevContainer
  - You can use Docker only, a `.devcontainer/Dockerfile` is provided
  - This video shows running the Docker container using VSCode DevContainers

CSci Operating Systems Class Resources

- `docs`: Open source `xv6-book.pdf` textbook with instructor highlights and notes
  - Also the `xv6-src-booklet-rev5.pdf` though page breaks are incorrect
- `slides`: slide decks and source file for lecture video slides
- `code`: source code examples used in lectures
  - See discussion in setup for how to use
- `xv6`: location where the `xv6-riscv` teaching os source code is placed

Setting up xv6 Source Code Compilation and Emulation

- Testing your installation
- Source code setup `make setup`
- Compiling and running xv6-riscv
- Running xv6 with gdb debugger
  - See the `gdb Debugging` lecture for additional help
- Using the lecture source code examples
  - And in general adding and compiling user programs

Summary

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References